

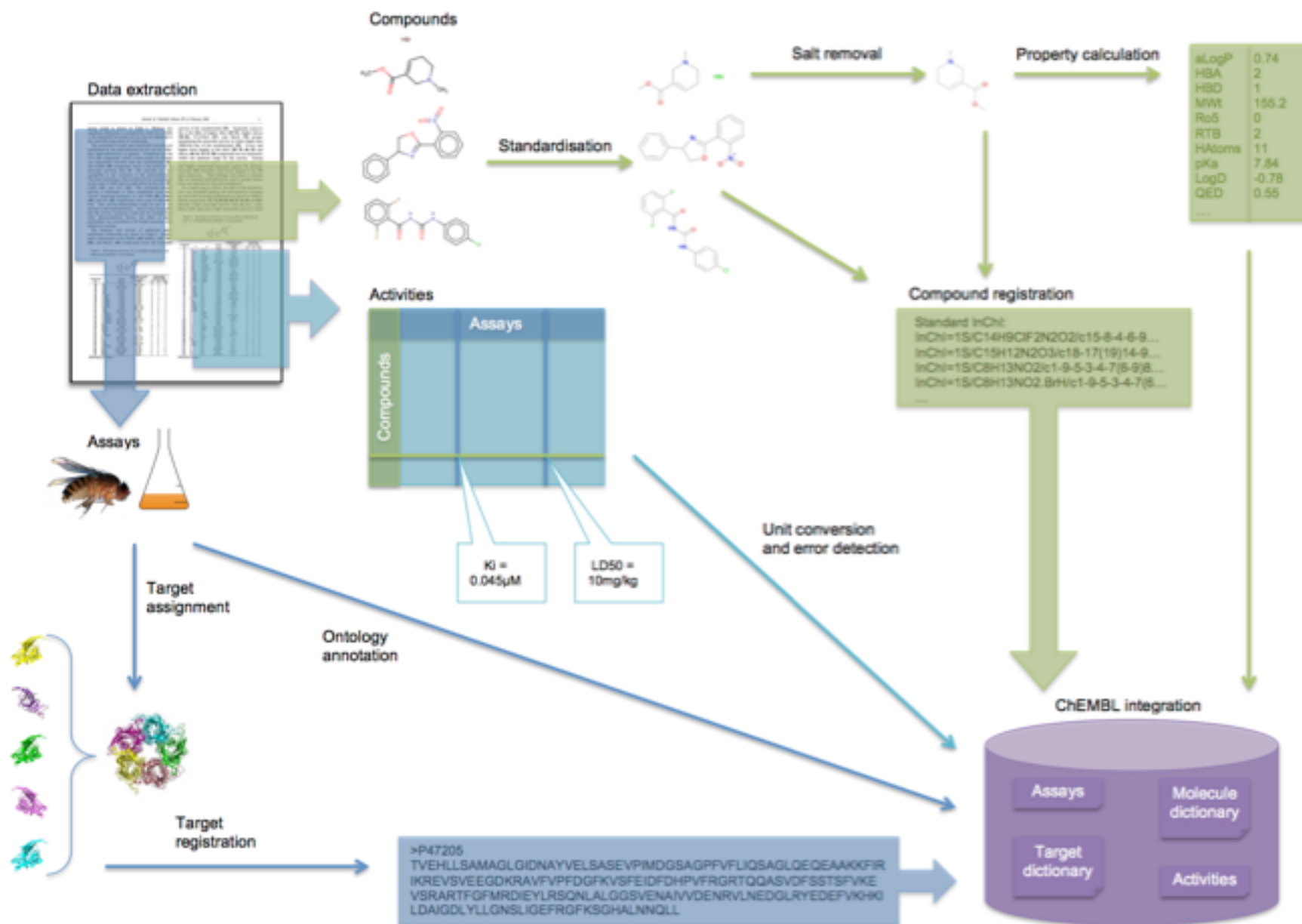
ChEMBL – data sharing and standards

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Chemogenomics Team

www.ebi.ac.uk

2. Organization, integration, curation and standardization of pharmacology data

ChEMBL Curation



Other Data Sources

BioAssay Data Sources:	Assays	Compounds	Activities
Scientific Literature	1,016,903	1,255,440	5,913,066
PubChem BioAssays	3,076	531,694	7,559,601
BindingDB Database	1,868	69,421	99,039
Patent Bioactivity Data	1,134	11,358	20,941
Sanger Institute Genomics of Drug Sensitivity i	714	139	73,169
GSK Published Kinase Inhibitor Set	456	1,101	169,451
Drugs for Neglected Diseases Initiative (DNDi)	233	7,070	14,452
MMV Malaria Box	138	8,438	45,158
MMV Pathogen Box	82	774	3,857
Open Source Malaria Screening	22	221	344
CO-ADD antimicrobial screening data	22	20	180
St Jude Malaria Screening	16	1,524	5,456
WHO-TDR Malaria Screening	16	740	5,853
AstraZeneca Deposited Data	15	5,799	11,687
GSK Tuberculosis Screening	15	826	1,814
Deposited Supplementary Bioactivity Data	13	1,786	4,817
GSK Kinetoplastid Screening	13	592	7,235
Curated Drug Metabolism Pathways	11	867	11
St Jude Leishmania Screening	6	13,643	42,105
GSK Malaria Screening	6	13,533	81,198
Novartis Malaria Screening	6	10,119	27,888
Harvard Malaria Screening	4	37	111
Gates Library compound collection	2	137,238	69,444
Open TG-GATEs	158,199	633	158,199
DrugMatrix	113,678	2,706	350,929
TP-search Transporter Database	3,592	4,383	6,765
FDA Approval Packages	1,386	80	1,387
Curated Drug Pharmacokinetic Data	521	98	1,163

2,101,843 compound records
 1,735,442 compounds
 14,675,320 activities
 1,302,147 assays
 11,538 targets
 67,722 documents

Compound-Only Data Sources:	Compounds
USP Dictionary of USAN and International Drug Names	11196
WHO Anatomical Therapeutic Chemical Classification	3099
Orange Book	2066
British National Formulary	1895
Clinical Candidates	1834
Gene Expression Atlas Compounds	793
Manually Added Drugs	392
Withdrawn Drugs	192
HeCaToS Compounds	96

Supplementary Bioactivity Data

- Rationale:
 - A paper contains enough data to support the “SAR story” but more data would have been generated in the course of the work
 - Scaffolds are disclosed so not of interest from an IP perspective
 - Why not share the related data?
- Data from GSK, Roche and University of Vienna on P2X7, motilin, ghrelin, Pgp, Carnitine palmitoyltransferases
- More details:
 - <http://chembl.blogspot.co.uk/2013/05/direct-submissions-of-data-to-chembl.html>

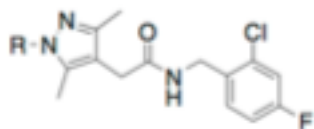
Supplementary Bioactivity Data

ABSTRACT

Example: Chambers et al, Bioorg. Med. Chem. Lett. (2010)

High-throughput screening identified compound **1** as a potent P2X₇ receptor antagonist suitable for lead optimisation. Structure-activity relationships (SAR) of a series of (1*H*-pyrazol-4-yl)acetamides were investigated and compound **32** was identified as a potent P2X₇ antagonist with enhanced potency and favourable physicochemical and pharmacokinetic properties.

Table 4
Structure-activity data for a range of N-substituents



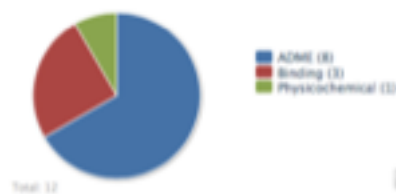
Compd	R	Human P2X ₇ pIC ₅₀ ^a	Rat Cl _{int} ^b mL/min/g liver
23		7.2	12.0
24		7.0	—
25		8.0	3.2
26		7.3	1.8
27		6.5	>50
28		6.9	20.0
29		6.5	5.7
30		6.7	2.4
31	Me	6.8	3.5
32	H	7.7	0.58

Table 5
Summary of the pharmacokinetic profile of compound **32**

Route of administration	Dose	Property	Measured value
IV ^a	1 mg/kg	CL _b (mL/min/kg)	15
		T _{1/2} (h)	1.7
		VD _{ss} (L/kg)	1.6
PO ^b	3 mg/kg	T _{max} (h)	1.0
		C _{max} (μM)	2.92 ± 0.33
		AUC/dose (min·kg/L)	58 ± 3.6
		F _{po}	90%

Assay Summary

CHEMBL Assays for Doc CHEMBL1157114

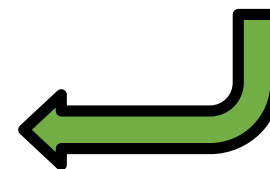


Bioactivity Summary

CHEMBL Activity Types for Doc CHEMBL1157114

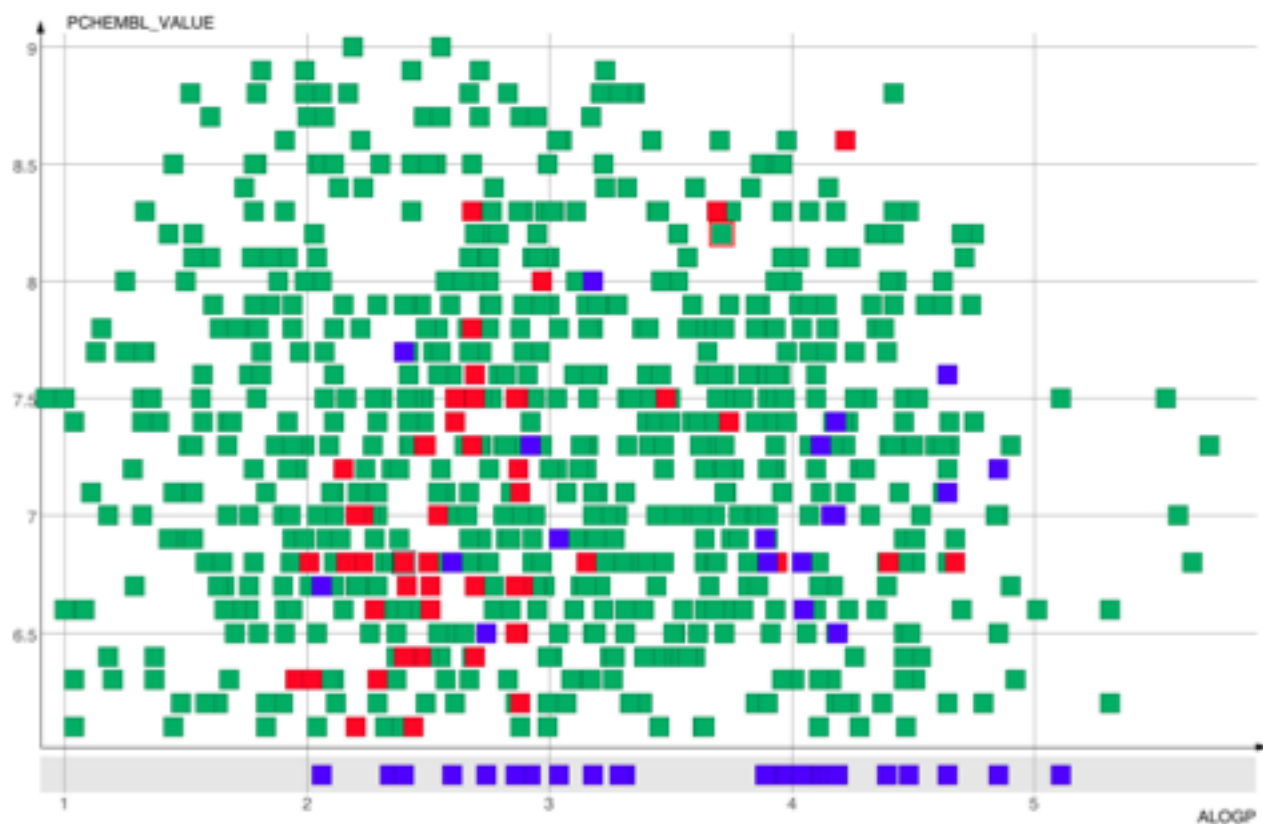


Data from article in ChEMBL



What does Supplementary Data add?

- Journal article 23 compounds active against P2X7 and 11 inactive ($\text{pIC}_{50} < 6$)
- Supp. Set 1 (ChEMBL2094195) – 54 new active compounds
- Supp. Set 2 (ChEMBL2218064) – 725 new active compounds



■ Original publication
■ Supp Set 1
■ Supp Set 2

DOC_CHEMBL_ID ■ CHEMBL1157114 ■ CHEMBL2094195 ■ CHEMBL2218064

ADME and Physchem Dataset

- Deposited by AstraZeneca
- Criteria for inclusion - compound was already in the public domain
- Identified compounds already in ChEMBL
- 11,687 activities on 5799 compounds

Assay	Compounds
Clint (Human hepatocytes)	408
Clint (Human liver microsomes)	1102
Clint (Rat hepatocytes)	837
Plasma Binding (Human)	1614
Plasma Binding (Rat)	717
Plasma Binding (dog)	244
Plasma Binding (Mouse)	162
Plasma Binding (Guinea Pig)	91
Solubility (pH7.4)	1763
LogD(pH7.4)	4200
pKa A1	221
pKa A2	25
pKa B1	261
pKa B2	40
pKa B3	2

Data standardisation

Standardise activity types
Including:
Antilog pKi, -logIC50 data etc

Standardise units

Flag data
Out of range
Incorrect units

Calculate a pCHEMBL value as:
-log₁₀ (molar IC50, XC50, EC50,
AC50, Ki, Kd or Potency)

Potential Duplicates

STANDARD_TYPE	PUBLISHED_ACTIVITY_TYPE
IC50	IC50_Mean
IC50	IC50_app
IC50	IC50_max
IC50	IC50
IC50	I 50
IC50	I-50
IC50	Average IC50
IC50	Mean IC50
IC50	CIC50
IC50	I50
IC50	IC50(app)

nM	pmol/ml	10 ⁻⁵ nM	M/dme3
nM	uM equiv	10 ⁻⁹ uM	10 ⁻⁶ uM
nM	pM ml ⁻¹	nmol/ml	umol
nM	10 ⁻⁶ mol/dm ³	10 ⁻² nM	10 ⁻¹ umol/L
nM	10 ⁻⁷ /M	10 ⁻⁵ mM	x10 ⁻¹⁰ M
nM	micromol/L	10 ⁻² mol/L	10 ⁻⁸ mM
nM	10 ⁻⁷ mol/L	uM ml ⁻¹	nm
nM	10 ⁻⁸ mol/L	mM	mol/L
nM	pmol/mL	10 ⁻⁴ mol/L	uM I-1
nM	10 ⁻¹ nM	10 ⁻⁴ M	10 ⁻⁶ mM
nM	10 ⁻⁵ nM	10 ⁻⁵ M	Nm
nM	nM ml ⁻¹	umol/L	10 ⁻³ microM
nM	10 ⁻⁵ nM	microM	10 ⁻⁶ nM
nM	10 ⁻² nM	10 ⁻⁶ mol/L	10 ⁻¹ uM
nM	10 ⁻⁹ mM	10 ⁻⁸ M	10 ⁻³ mmol/L
nM	10 ⁻⁶ /M	10 ⁻⁸ M	uM/L
nM	nmol/l	10 ⁻¹¹ M	um
nM	NM	10 ⁻² nmol	nM/ml
nM	M,mol dme-3	10 ⁻³ nmol	10 ⁻³ M
nM	10 ⁻⁵ nM	10 ⁻⁵ /M	10 ⁻⁵ M

Data standardisation

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PUBLISHED_TYPE	PUBLISHED_VALUE	PUBLISHED_UNITS	DATA_VALIDITY_COMMENT
IC50	0.0022	ml	Non standard unit for type
IC50	0.005	ug	Non standard unit for type
IC50	7.4	M	Outside typical range
IC50	0.000000038	nM	Outside typical range

Calculate a pCHEMBL value as:
-log₁₀ (molar IC50, XC50, EC50,
AC50, Ki, Kd or Potency)

STANDARD_TYPE	STANDARD_VALUE	STANDARD_UNITS	PCHEMBL_VALUE
IC50	565	nM	6.25
IC50	61000	nM	4.21
Ki	140	nM	6.85
Ki	168	nM	6.77
IC50	0.3	nM	9.52

Potential Duplicates

Data standardisation

Standardise activity types
Including:
Antilog pKi, -logIC50 data etc

Standardise units

Flag data
Out of range
Incorrect units

Calculate a pCHEMBL value as:
-log₁₀ (molar IC₅₀, XC₅₀, EC₅₀,
AC₅₀, Ki, K_d or Potency)

Potential Duplicates

MOLREGNO	PREF_NAME	STANDARD_TYPE	STANDARD_VALUE	STANDARD_UNITS	PCHEMBL_VALUE	JOURNAL	YEAR	POTENTIAL_DUPLICATE
34997	Protein kinase C delta	Ki	66.2	nM	7.18	J. Med. Chem.	1999	
34997	Protein kinase C delta	Ki	66.2	nM	7.18	J. Med. Chem.	2001	1

Description of assays

- Early use-cases for ChEMBL:
 - ‘find compounds that are active against this target’
 - ‘find all of the targets that my compound is active against’
- Primarily rely on Compound – IC50 value – Target queries
- Assays assigned a textual description based on paper e.g.,
 - *Tested in vitro for the inhibition of [3H]8-OH-DPAT binding to 5-hydroxytryptamine 1A receptor, expressed in cloned CHO cells.*
- Compound and target information recorded in more detail e.g., salt form of compound, type of target (single protein vs protein complex)
- Needed to revisit publication for more details of assays

Pooling data from different assays/publications

- When pooling or comparing data from different publications/assays, the combination of compound/target/activity value usually not sufficient
- Info such as cell line, radioligand/substrate etc important

molecular
informatics

DOI: 10.1002/minf.201200059

Annotating Human P-Glycoprotein Bioassay Data

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Abstract: Huge amounts of small compound bioactivity data have been entering the public domain as a consequence of open innovation initiatives. It is now the time to carefully analyse existing bioassay data and give it a systematic structure. Our study aims to annotate prominent in vitro assays used for the determination of bioactivities of human P-glycoprotein inhibitors and substrates as they are represented in the ChEMBL and TP-search open source databases. Furthermore, the ability of data, determined in different assays, to be combined with each other is explored.

As a result of this study, it is suggested that for inhibitors of human P-glycoprotein it is possible to combine data coming from the same assay type, if the cell lines used are also identical and the fluorescent or radiolabeled substrate have overlapping binding sites. In addition, it demonstrates that there is a need for larger chemical diverse datasets that have been measured in a panel of different assays. This would certainly alleviate the search for other inter-correlations between bioactivity data yielded by different assay setups.

Keywords: P-glycoprotein • Multidrug resistance protein 1 • Bioassay ontology • ChEMBL database • TP-search

SPECIAL ISSUE

OPINION

Minimum information about a bioactive entity (MIABE)

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Abstract | Bioactive molecules such as drugs, pesticides and food additives are produced in large numbers by many commercial and academic groups around the world. Enormous quantities of data are generated on the biological properties and quality of these molecules. Access to such data — both on licensed and commercially available compounds, and also on those that fail during development — is crucial for understanding how improved molecules could be developed. For example, computational analysis of aggregated data on molecules that are investigated in drug discovery programmes has led to a greater understanding of the properties of successful drugs. However, the information required to perform these analyses is rarely published, and when it is made available it is often missing crucial data or is in a format that is inappropriate for efficient data-mining. Here, we propose a solution: the definition of reporting guidelines for bioactive entities — the Minimum Information About a Bioactive Entity (MIABE) — which has been developed by representatives of pharmaceutical companies, data resource providers and academic groups.

The process of identifying and developing molecules with useful bioactive properties, such as pharmaceutical agents, pesticides and food additives, is fraught with difficulty. In the pharmaceutical industry only a very small percentage of investigational new drugs will make it through to approved clinical usage, for reasons that include lack of efficacy, unexpected side effects and toxicity, and undesirable drug–drug interactions. Active pesticides that have undesirable side effects against organisms other than their original target will also not make it to the market. Food additives such as plant sterols are subject to strict regulations on their safety and possible off-target effects.

Published reports on the properties of ‘failed’ molecules, in addition to detailed

information on those that become fully licensed and commercially available, are crucial for developing an understanding of how improved molecules may be developed. Details of the molecular structure and mechanism of action of molecules may give clues as to how related analogues could be developed that modulate the same target, but with improved effectiveness or increased specificity. A full disclosure of observed toxicity or an understanding of the pharmacokinetic properties of an agent may help to improve these properties in subsequent generations of molecules. Even those molecules that fail at an early stage in the development process may be useful as tools; for example, in aiding the validation of potential new targets

that are identified through genomic and/or proteomic studies of disease.

Another area in which integrating knowledge on bioactive entities could be valuable is in identifying proteins that could be amenable to modulation by small molecules. In 2002, Hopkins and Groom¹ introduced the concept of the ‘druggable genome’ to describe the portion of the human genome that encoded proteins that were both linked to disease and were tractable targets for small-molecule drugs. Since then, progress has been made with computational approaches to calculate the druggability of proteins and to predict druggable proteins^{2–4}. In 2006, Overington et al.⁵ estimated that the number of protein targets for approved drugs at the time was as low as 266 out of the 20,400 protein-coding genes⁶ in the human genome (a further 58 targets were either from pathogenic organisms or they were non-protein molecules). Similar models may be developed to assess the susceptibility of proteins encoded by plant or microbial genomes to modulation by small molecules. Mining computationally accessible resources of data on bioactive entities could provide synergistic benefits for research and development across the fields of drug discovery, pesticide science, cosmetics, nutraceuticals, metabolomics and toxicology.

At present, however, much of the substantial volume of data that would aid predictions of protein targets that could be successfully modulated, and the development of new or improved molecules to modulate these targets, resides in proprietary databases. This is especially the case for agents that have subsequently failed at some stage of the discovery process (for example, owing to poor pharmacokinetics or toxicity). Nevertheless, owing to the current productivity crisis and decreasing budgets for early-stage research, the pharmaceutical, biotechnology and food industries are increasingly considering the precompetitive release of compound-related bioactivity data into public repositories, recognizing that this could not only provide a general benefit for research but also a commercial advantage as data only need to be collected once. The manual curation of data is an expensive process and requires resources that are often not available in even the largest pharmaceutical

Ontology annotation

- Possible to retrospectively annotate assay descriptions with ontology terms e.g.,
 - BioAssay Ontology
 - Uberon
 - Cell-Line Ontology
- Allows more sophisticated queries and more robust comparisons e.g.,



Assay Parameters

- While some parameters can be annotated from assay descriptions, others may not be captured e.g.,
 - *Tested in vitro for the inhibition of [3H]8-OH-DPAT binding to 5-hydroxytryptamine 1A receptor, expressed in cloned CHO cells.*
- Needed a place to store these additional parameters where we have them

Assay Parameters

Parameter	Value
ASSAY_TEST	Inhibition
CONC	= 80 nM
DATASET	Absorbance
REAGENT_READOUT	Biomol Green
STUDY_ID	Ld-MetRS
TIME	= 2.5 hr

In vivo/toxicity data

- ChEMBL now incorporating more complex types of data e.g., in vivo pathology measurements, drug pharmacokinetic data
- Parameters may be different for different activity measurements (e.g., compound dose or time point of measurement) – assay parameters not sufficient
- Activity properties:

Property	Value
DOSAGE_FORM	Gel
DOSE	= 0.25g
N	= 38
REGIMEN	Multiple daily dose
ROUTE	Topical
SAMPLING_TIME	Day 14
TISSUE	Plasma

- Allows more flexible data deposition



Updated format
for assay
parameters (to
match other
tables)

Summary

- Important to define minimum information requirements
 - Hard to go back later (e.g., to full text literature) and capture more information
 - But difficult to anticipate all use-cases up front (different use-cases may have different requirements)
 - Differences between summary data from e.g., literature and data archiving (deposited data sets)
- Data standardisation/vocabularies also important, but can be possible to standardise retrospectively
- Flexible model can allow more information to be captured but can be harder to standardise

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